5.25PT 汇报

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Section 1

T6 Non-contact Resistance

The responses of an LRC circuit driven by an AC source can be changed by inserting either a non-magnetic metal rod or a ferromagnetic rod into the inductor coil. How can we obtain the magnetic and electric properties of the inserted rod from the circuit's responses?

场方程试求解

Field Eq.

$$\nabla^2 \vec{B} + \epsilon\mu \frac{\partial^2 \vec{B}}{\partial t^2} + \mu\sigma \frac{\partial \vec{B}}{\partial t} = 0$$

由于 $\vec{B} = \vec{B_0} + \vec{B'}$，而 $\vec{B_0} = B_0 \sin(\omega t + \phi)\vec{e_z}$，$\vec{B'} = B'(r,t)\vec{e_z}$ 代入有

$$\nabla^2 \vec{B'} + \epsilon\mu \frac{\partial^2 \vec{B'}}{\partial t^2} + \mu\sigma \frac{\partial \vec{B'}}{\partial t} = \left(\epsilon\mu\omega^2 B_0 \sin(\omega t + \phi) - \mu\sigma\omega B_0 \cos(\omega t + \phi)\right) \vec{e_z}$$

显然，这里将 $\vec{B'}$ 按柱坐标展开求解

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial B'}{\partial r}\right) + \epsilon\mu \frac{\partial^2 B'}{\partial t^2} + \mu\sigma \frac{\partial B'}{\partial t} = f(t)$$

其中 $f(t) = \epsilon\mu\omega^2 B_0 \sin(\omega t + \phi) - \mu\sigma\omega B_0 \cos(\omega t + \phi)$ 是系统的驱动项，显然这是一个非齐次线性偏微分方程。

为了求解，我们设线性微分算子

$$\hat{L} \equiv \hat{R} + \hat{T} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \epsilon\mu \frac{\partial^2}{\partial t^2} + \mu\sigma \frac{\partial}{\partial t}$$

显然，原方程化为

$$\hat{L}B'(r,t) = \hat{L}u(r,t) + \hat{T}v(t) = f(t)$$

场方程试求解

对于非齐次项，即为

$$\hat{T}v(t) = f(t)$$

我们采用格林函数法求解，先求出其齐次方程的解

$$\hat{T}\psi(t) = 0$$

$$\hat{T} = \epsilon\mu\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \mu\sigma\frac{\mathrm{d}}{\mathrm{d}t}$$

我们令 $\psi(t) = \mathrm{e}^{-\mathrm{i}\omega' t}$，带入后得到特征方程

$$\epsilon\mu\omega'^2 + \mathrm{i}\mu\sigma\omega' = 0$$

则 $\psi_1(t) = \mathrm{e}^{-\frac{\sigma}{\epsilon}t}, \psi_2(t) = 1$。

我们再求对应的格林函数。定义格林函数 $G(t, t')$ 满足

$$\hat{T}G(t, t') = \delta(t - t') \quad t, t' > 0$$

则初值条件有

$$G(0, t') = 0, \quad \left[\frac{\mathrm{d}}{\mathrm{d}t}G(t, t')\right]_{t=0} = 0$$

求解得

$$G(t, t') = \frac{\mathrm{e}^{-\frac{\sigma}{2\epsilon}(t-t')}}{\mathrm{i}\frac{\sigma}{2\epsilon}}\sin\left[\mathrm{i}\frac{\sigma}{2\epsilon}(t - t')\right]\theta(t - t')$$

场方程试求解

由上页的求解，我们可以得到非齐次解的结果

$$
\begin{aligned}
v(t) = & C_1 \mathrm{e}^{-\frac{\sigma}{\epsilon}t} + C_2 + \\
& \int_0^\infty \mathrm{d}t' \left\{ \frac{\mathrm{e}^{-\frac{\sigma}{2\epsilon}(t-t')}}{\mathrm{i}\frac{\sigma}{2\epsilon}} \sin\left[\mathrm{i}\frac{\sigma}{2\epsilon}(t-t')\right] \theta(t-t') \right. \\
& \left. \left[\omega^2 B_0 \sin(\omega t' + \phi) - \frac{\sigma}{\epsilon}\omega B_0 \cos(\omega t' + \phi)\right] \right\}
\end{aligned}
$$

场方程试求解

对于齐次项，我们可以通过分离变量法求解得到。

设 $u(r,t) = R(r)\,T(t)$，代入齐次方程分离变量后有

$$\frac{\hat{R}R(r)}{R(r)} = -\frac{\hat{T}T(t)}{T(t)} = -\lambda^2$$

对 $R(r)$，化简得到

$$r^2\frac{\mathrm{d}^2R}{\mathrm{d}r^2} + r\frac{\mathrm{d}R}{\mathrm{d}r} + \lambda^2 r^2 R = 0$$

令 $x = \lambda r$，则方程化为

$$x^2\frac{\mathrm{d}^2R}{\mathrm{d}x^2} + x\frac{\mathrm{d}R}{\mathrm{d}x} + x^2 R = 0$$

显然，这是一个 0 阶贝塞尔方程，求解有

$$R(x) = C_3\,J_0(x) + C_4\,Y_0(x)$$

其中

$$J_0(x) = \sum_{k=0}^{\infty}\frac{(-1)^k}{(m!)^2}\left(\frac{x}{2}\right)^{2k}$$

$$Y_0(x) = \frac{2}{\pi}J_0(x)\ln\frac{x}{2} - \frac{2}{\pi}\sum_{k=0}^{\infty}\frac{(-1)^k\psi(k+1)}{(m!)^2}\left(\frac{x}{2}\right)^{2k}$$

场方程试求解

对 $T(t)$，化简得到

$$(\hat{T} - \lambda^2)\, T(t) = 0$$

我们令 $T(t) = \mathrm{e}^{-\mathrm{i}\omega'' t}$，带入后得到特征方程

$$\epsilon\mu\omega''^2 + \mathrm{i}\mu\sigma\omega'' + \lambda^2 = 0$$

解得

$$\omega_1'' = \mathrm{i}\frac{1}{2\epsilon}\left(-\sigma + \sqrt{\sigma^2 + 4\lambda^2\frac{\epsilon}{\mu}}\right)$$

$$\omega_2'' = \mathrm{i}\frac{1}{2\epsilon}\left(-\sigma - \sqrt{\sigma^2 + 4\lambda^2\frac{\epsilon}{\mu}}\right)$$

则解得

$$T(t) = C_5\mathrm{e}^{-\mathrm{i}\omega_1'' t} + C_6\mathrm{e}^{-\mathrm{i}\omega_2'' t}$$

场方程试求解

综上所述，我们可以得到

$$
\begin{aligned}
B(r,t) =& B_0(t) + B'(r,t) = B_0(t) + u(r,t) + v(t) = B_0(t) + R(r)\,T(t) + v(t) \\
=& B_0 \sin(\omega t + \phi) + \left[C_3 J_0(\lambda r) + C_4 Y_0(\lambda r)\right] \left[C_5 \mathrm{e}^{-\mathrm{i}\omega_1'' t} + C_6 \mathrm{e}^{-\mathrm{i}\omega_2'' t}\right] + \\
& C_1 \mathrm{e}^{-\frac{\sigma}{\epsilon} t} + C_2 + \int_0^{\infty} \mathrm{d}t' \left\{ \frac{\mathrm{e}^{-\frac{\sigma}{2\epsilon}(t-t')}}{\mathrm{i}\frac{\sigma}{2\epsilon}} \sin\left[\mathrm{i}\frac{\sigma}{2\epsilon}(t-t')\right] \theta(t-t') \right. \\
& \left. \left[\omega^2 B_0 \sin(\omega t' + \phi) - \frac{\sigma}{\epsilon}\omega B_0 \cos(\omega t' + \phi)\right] \right\}
\end{aligned}
$$

接下来的任务：

1. 确定合适的边界条件和初值条件，来确定常数。
2. 尝试用 matlab 作图表示成直观的形式。
3. 极限分析，观察稳态解。
4. 与实验值做对比。

Section 2

T14 Ruler Trick

Place a ruler on the edge of a table, and throw a ball at its free end. The ruler will fall. However, if you cover a part of the ruler with a piece of paper and repeat the throw, then the ruler will remain on the table while the ball will bounce off it. Explain this phenomenon, and investigate the relevant parameters.